

Angles, Polygons & Parallel Lines

Test · 28 marks total

Name: _____

Date: _____

Score: _____ / 28

Section A — Angle facts

A1. Two angles lie on a straight line. One angle is 118° . Find the other angle.

[1 mark]

A2. Three angles meet at a point: 95° , 140° and x° . Find x .

[1 mark]

A3. Two straight lines cross. One angle is 42° . State the size of the angle vertically opposite to it, giving a reason.

[2 marks]

A4. Two angles on a straight line are $(4x)^\circ$ and $(2x + 30)^\circ$. Find the value of x .

[2 marks]

Section B — Triangles & quadrilaterals

B1. A triangle has angles 48° , 76° and x° . Find x .

[2 marks]

B2. An isosceles triangle has a base angle of 70° . Find the other two angles of the triangle.

[2 marks]

B3. A quadrilateral has angles 90° , 105° , 88° and x° . Find x .

[2 marks]

B4. A kite has angles 100° , 70° , x° and x° (the last two angles are equal). Find x .

[2 marks]

Section C — Parallel lines

C1. What name is given to a pair of angles that form a "Z" shape between two parallel lines?

[1 mark]

C2. One angle is 108° . Find the size of its corresponding angle.

[1 mark]

C3. One angle is 62° . Find the size of its co-interior (allied) angle.

[1 mark]

C4. Two parallel lines are cut by a transversal. One angle is $(2x + 15)^\circ$, and its corresponding angle is 85° . Find the value of x .

[3 marks]

Section D — Polygons

D1. Calculate the sum of the interior angles of a hexagon.

[2 marks]

D2. A regular polygon has 12 sides. Calculate the size of each exterior angle.

[2 marks]

D3. Calculate the size of each interior angle of a regular decagon (10 sides).

[2 marks]

D4. A regular polygon has an interior angle of 150° . How many sides does it have?

[2 marks]